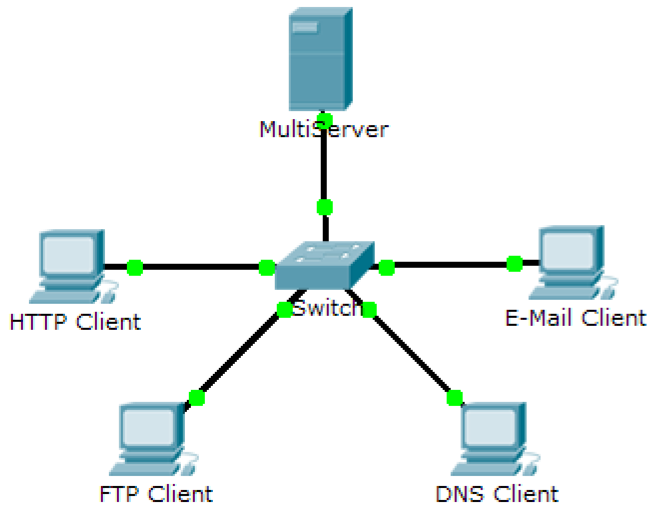
Packet Tracer Simulation - TCP and UDP Communications

Topology



Objectives

Part 1: Generate Network Traffic in Simulation Mode

Part 2: Examine the Functionality of the TCP and UDP Protocols

Background

This simulation activity is intended to provide a foundation for understanding the TCP and UDP in detail. Simulation mode provides the ability to view the functionality of the different protocols.

As data moves through the network, it is broken down into smaller pieces and identified in some fashion so that the pieces can be put back together. Each of these pieces is assigned a specific name (protocol data unit [PDU]) and associated with a specific layer. Packet Tracer Simulation mode enables the user to view each of the protocols and the associated PDU. The steps outlined below lead the user through the process of requesting services using various applications available on a client PC.

This activity provides an opportunity to explore the functionality of the TCP and UDP protocols, multiplexing and the function of port numbers in determining which local application requested the data or is sending the data.

Part 1: Generate Network Traffic in Simulation Mode

Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

a. Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.

b. Enter the **ping 192.168.1.255** command. This will take a few seconds as every device on the network responds to **MultiServer**.

c. Close the **MultiServer** window.

Step 2: Generate web (HTTP) traffic.

a. Switch to Simulation mode.

b. Click **HTTP Client** and click the **Desktop** tab > **Web Browser**.

c. In the URL field, enter **192.168.1.254** and click **Go**. Envelopes (PDUs) will appear in the simulation window.

d. Minimize, but do not close, the **HTTP Client** configuration window.

Step 3: Generate FTP traffic.

a. Click **FTP Client** and click the **Desktop** tab > **Command Prompt**.

b. Enter the **ftp 192.168.1.254** command. PDUs will appear in the simulation window.

c. Minimize, but do not close, the **FTP Client** configuration window.

Step 4: Generate DNS traffic.

a. Click **DNS Client** and click the **Desktop** tab > **Command Prompt**.

b. Enter the **nslookup multiserver.pt.ptu** command. A PDU will appear in the simulation window.

c. Minimize, but do not close, the **DNS Client** configuration window.

Step 5: Generate Email traffic.

a. Click **E-Mail Client** and click the **Desktop** tab > **E Mail** tool.

b. Click **Compose** and enter the following information:

   1) **To:** user@multiserver.pt.ptu

   2) **Subject:** Personalize the subject line

   3) **E-Mail Body:** Personalize the Email

c. Click **Send**.

d. Minimize, but do not close, the **E-Mail Client** configuration window.

Step 6: Verify that the traffic is generated and ready for simulation.

Every client computer should have PDUs listed in the Simulation Panel.

Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine multiplexing as all of the traffic crosses the network.

You will now use the **Capture/Forward button** and the **Back** button in the Simulation Panel.

a. Click **Capture/Forward** once. All of the PDUs are transferred to the switch.

b. Click **Capture/Forward** again. Some of the PDUs disappear. What do you think happened to them?
The switch forwards them to their respective destinations. This action reflects the process of multiplexing where multiple protocols (HTTP, FTP, DNS, etc.) are combined and transmitted over the network simultaneously using different port numbers. The switch determines the correct destination for each packet and forwards it accordingly.

c. Click **Capture/Forward** six times. All clients should have received a reply. Note that only one PDU can cross a wire in each direction at any given time. What is this called?
This concept is called serialization which is the sequential transmission of packets over a medium where only one packet is sent across a link at a time in a specific direction.

d. A variety of PDUs appears in the event list in the upper right pane of the simulation window. Why are they so many different colors?
The variety of colors reflects the different protocols and layers involved in the end-to-end communication process because each protocol serves a specific function in the transmission of data, and Packet Tracer uses these colors to show how data flows through the layers of the OSI or TCP/IP model, helping users identify and analyze the traffic more effectively.

The color green is utilized to represent Application Layer Protocols such as HTTP, DNS, or SMTP traffic. These represent user-level services like web browsing, email, and domain name resolution.

The color purple is selected to represent Transport Layer Protocols such as TCP or UDP. This shows how the data is broken into segments, assigned port numbers, and managed for delivery reliability (TCP) or speed (UDP).

The color brown is used to represent Network Layer Protocols such as IP. These PDUs handle addressing and routing of packets between source and destination.

The color grey is chosen to visualize Data Link Layer Protocols such as Ethernet frames. These PDUs manage communication on the physical network, including MAC addressing and frame delivery.

e. Click **Back** eight times. This should reset the simulation.

Note: Do not click **Reset Simulation** any time during this activity; if you do, you will need to repeat the steps in Part 1.

Step 2: Examine HTTP traffic as the clients communicate with the server.

a. Filter the traffic that is currently displayed to display only **HTTP** and **TCP** PDUs filter the traffic that is currently displayed:

1) Click **Edit Filters** and toggle the **Show All/None** check box.

2) Select **HTTP** and **TCP**. Click anywhere outside of the Edit Filters box to hide it. The Visible Events should now display only **HTTP** and **TCP** PDUs.

b. Click **Capture/Forward**. Hold your mouse above each PDU until you find one that originates from **HTTP Client**. Click the PDU envelope to open it.

c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?
TCP

Are these communications considered to be reliable?
Yes.These communications are considered reliable because the TCP section includes information about sequence numbers, acknowledgment numbers, and flags (like SYN, ACK, and FIN). TCP uses checksums to verify that the data is received correctly. The sender receives confirmation (ACK) for delivered data. If a packet is lost or corrupted, TCP will retransmit it.TCP also ensures data is delivered in the correct order using sequence numbers. Thus, HTTP communications rely on TCP for dependable and ordered delivery of data, making the communication reliable.

d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?
SOURE PORT:1029
DESTINATION PORT:80
SEQUENCE NUM:0
ACK NUM:0
The field to the left of the **WINDOW** field is FLAGS with the value of 0b00000010

e. Close the PDU and click **Capture/Forward** until a PDU returns to the **HTTP Client** with a checkmark.

f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

The SRC PORT and DEST PORT are swapped because the communication direction changes since the server is responding to client.

The Sequence Number remains 0 because the packet observed might still belong to the handshake process, where no payload data is being sent yet. The increment in the sequence number only happens after the initial data transmission.

The acknowledgment number increases to confirm receipt of the previous data. Since the client sent data with SEQUENCE NUM = 0, the server's acknowledgment might now show ACK NUM = 1.

g. There is a second **PDU** of a different color, which **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the HTTP communication. Click this second PDU envelope and select **Outbound PDU Details**.

h. What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?

Source Port: 1029 (same as before, still identifying the client-side port).

Destination Port: 80 (the server's port for HTTP traffic).

Sequence Number: 1 (indicating that this is the first segment of actual data being sent after the handshake).

Acknowledgment Number: 1 (confirming receipt of the server's initial segment during the handshake).

Differences from the previous two PDUs:

Sequence Number:

Initially, the sequence number was 0 during the handshake. Now it has increased to 1, indicating the start of actual HTTP data transmission.

Acknowledgment Number:

Initially, the acknowledgment number in the first client-server handshake packets was 0. It increased to 1 after the acknowledgment of the handshake, and it remains 1 here. This marks the transition from the TCP handshake phase to actual HTTP communication, where data begins flowing.

i. Click **Back** until the simulation is reset.

Step 3: Examine FTP traffic as the clients communicate with the server.

a. In the Simulation Panel, change **Edit Filters** to display only **FTP** and **TCP**.

b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **FTP Client**. Click that PDU envelope to open it.

c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

TCP

Are these communications considered to be reliable?

Yes.These communications are considered reliable because the TCP section includes information about sequence numbers, acknowledgment numbers, and flags (like SYN, ACK, and FIN). TCP uses checksums to verify that the data is received correctly. The sender receives confirmation (ACK) for delivered data. If a packet is lost or corrupted, TCP will retransmit it.TCP also ensures data is delivered in the correct order using sequence numbers. Thus, HTTP communications rely on TCP for dependable and ordered delivery of data, making the communication reliable.

d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

SOURE PORT:1025
DESTINATION PORT:21
SEQUENCE NUM:0
ACK NUM:0

The field to the left of the **WINDOW** field is FLAGS with the value of 0b00000010

e. Close the PDU and click **Capture/Forward** until a PDU returns to the **FTP Client** with a checkmark.

f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

The SRC PORT and DEST PORT are swapped because the communication direction changes since the server is responding to client.

The Sequence Number remains 0 because the packet observed might still belong to the handshake process, where no payload data is being sent yet. The increment in the sequence number only happens after the initial data transmission.

The acknowledgment number increases to confirm receipt of the previous data. Since the client sent data with SEQUENCE NUM = 0, the server's acknowledgment shows ACK NUM = 1.

g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

Source and Destination Ports:

In all PDUs, the **Source Port** remains **1025** (the client's dynamically assigned port), and the **Destination Port** remains **21** (the FTP server's control port). These ports identify the client-server connection for FTP communication. No changes were observed in the port values.

Sequence Number:

In earlier PDUs, The **Sequence Number** was **0**, as the communication was in the handshake phase, where no payload data had been sent yet.

In the current PDU, The **Sequence Number** has incremented to **1**, indicating the client has initiated data transmission as part of the FTP process. This number ensures the proper ordering of data packets during transmission.

Acknowledgment Number:

In earlier PDUs, the **Acknowledgment Number** was **0**, as no data from the server needed to be acknowledged at that stage.

In the current PDU, the **Acknowledgment Number** is **1**, confirming that the client has successfully received the data from the server with **Sequence Number = 0**.

Flags Field:

In earlier PDUs, the **Flags** field included values for handshake operations (e.g., SYN, SYN-ACK).
In the current PDU, the **Flags** field has the value **0b00010000**, which corresponds to the **ACK** flag. This indicates that the client acknowledges the receipt of data from the server.

h. Close the PDU and click **Capture/Forward** until a second PDU returns to the **FTP Client**. The PDU is a different color.

i. Open the PDU and select **Inbound PDU Details**. Scroll down past the TCP section. What is the message from the server?

The message from the server is a FTP Response with 2 fields which includes Code with value of 220 and Message with "Welcome to PT Ftp server".

j. Click **Back** until the simulation is reset.

Step 4: Examine DNS traffic as the clients communicate with the server.

a. In the Simulation Panel, change **Edit Filters** to display only **DNS** and **UDP**.

b. Click the PDU envelope to open it.

c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?
DNS Query

Are these communications considered to be reliable?

No, these communications are not considered reliable because DNS uses UDP (User Datagram Protocol), which is a connectionless protocol. Unlike TCP, UDP does not include mechanisms for ensuring data delivery, retransmission, or ordering. While UDP is faster due to its lightweight nature, it does not guarantee reliability. DNS relies on the application layer to handle potential issues like retransmissions or timeouts if a query fails.

d. Record the **SRC PORT** and **DEST PORT** values. Why is there no sequence and acknowledgement number?
SOURE PORT:1025
DESTINATION PORT:53

There is no sequence number or acknowledgment number because UDP is a stateless, connectionless protocol that does not establish a session or track the order of data packets. Unlike TCP, UDP does not need these fields since it does not guarantee delivery or order.

e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **DNS Client** with a checkmark.

f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?
The SRC PORT:53 and DEST PORT:1025 are swapped because the communication direction changes since the server is responding to client.
There is no sequence number or acknowledgment number because UDP is a stateless, connectionless protocol

g. What is the last section of the **PDU** called?
DNS Answer

h. Click **Back** until the simulation is reset.

Step 5: Examine email traffic as the clients communicate with the server.

a. In the Simulation Panel, change **Edit Filters** to display only **POP3, SMTP** and **TCP**.

b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **E-mail Client**. Click that PDU envelope to open it.

c. Click the **Inbound PDU Details** tab and scroll down to the last section. What transport layer protocol does email traffic use?
TCP

Are these communications considered to be reliable?
Yes.These communications are considered reliable because the TCP section includes information about sequence numbers, acknowledgment numbers, and flags (like SYN, ACK, and FIN). TCP uses checksums to verify that the data is received correctly. The sender receives confirmation (ACK) for delivered data. If a packet is lost or corrupted, TCP will retransmit it.TCP also ensures data is delivered in the correct order using sequence numbers. Thus, HTTP communications rely on TCP for dependable and ordered delivery of data, making the communication reliable.

d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?
SOURE PORT:1025
DESTINATION PORT:25
SEQUENCE NUM:0
ACK NUM:0

The field to the left of the WINDOW field is FLAGS with the value of 0b00000010

e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **E-Mail Client** with a checkmark.

f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?
The SRC PORT and DEST PORT are swapped because the communication direction changes since the server is responding to client.

The Sequence Number remains 0 because the packet observed might still belong to the handshake process, where no payload data is being sent yet. The increment in the sequence number only happens after the initial data transmission.

The acknowledgment number increases to confirm receipt of the previous data. Since the client sent data with SEQUENCE NUM = 0, the server's acknowledgment shows ACK NUM = 1.

g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?
SOURCE PORT:1025
DESTINATION PORT:25
SEQUENCE NUMBER:1
ACKNOWLEDGEMENT NUMBER:1

The SOURCE PORT and DESTINATION PORT remain the same as in earlier steps.
The SEQUENCE NUMBER increases to 1 because data is now being sent.
The ACKNOWLEDGMENT NUMBER remains 1 to confirm receipt of the server's data from the previous step.

h. There is a second **PDU** of a different color that **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the email communication. Click this second PDU envelope and select **Outbound PDU Details.**

i. How are the port and sequence numbers different from the previous two **PDU**s?

SOURCE PORT:1025
DESTINATION PORT:25
SEQUENCE NUMBER:1
ACKNOWLEDGEMENT NUMBER:1

The SOURCE PORT and DESTINATION PORT remain consistent, as the communication still occurs between the client (port 1025) and the server (port 25).
The SEQUENCE NUMBER remains 1 because this packet is likely continuing the current session and no additional data has been transmitted since the handshake.
The ACKNOWLEDGMENT NUMBER also remains 1, confirming receipt of the previously transmitted data.

j. What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?

TCP Port 25: The email protocol associated with TCP port 25 is SMTP (Simple Mail Transfer Protocol). SMTP is used for sending emails from clients to servers and between mail servers.

TCP Port 110: The email protocol associated with TCP port 110 is POP3 (Post Office Protocol version 3). POP3 is used for retrieving emails from a mail server to a client.

k. Click **Back** until the simulation is reset.

Step 6: Examine the use of port numbers from the server.

a. To see TCP active sessions, perform the following steps in quick succession:

1) Switch back to **Realtime** mode.

2) Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.

b. Enter the **netstat** command. What protocols are listed in the left column?

What port numbers are being used by the server?

TCP

Local Address: 192.168.1.254:21

Foreign Address: 192.168.1.2:1025

The **:21** in the local address corresponds to the **FTP** service.

Port **1025** in the foreign address is a dynamically assigned client port used for this specific session.

c. What states are the sessions in?

ESTABLISHED

d. Repeat the **netstat** command several times until you see only one session still ESTABLISHED. For which service is this connection still open?

The connection still open is for the FTP (File Transfer Protocol) service.

Why doesn’t this session close like the other three? (Hint: Check the minimized clients)

The session remains open because the **FTP client** is still active and waiting for additional user commands. Unlike other services, FTP often requires multiple interactions between the client and the server during an ongoing session (e.g., commands like LIST, RETR, or STOR), so the session does not close immediately.

Suggested Scoring Rubric

Activity Section	Question Location	Possible Points	Earned Points
Part 2: Examine Functionality of the TCP and UDP Protocols	Step 1	15	
	Step 2	15	
	Step 3	15	
	Step 4	15	
	Step 5	15	
	Step 6	25	
	Total Score	**100**	